Quotient-Ring Trigonometry

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Abstract

This paper presents a general framework for constructing planar trigonometries on curves defined by arbitrary homogeneous polynomials, extending the classical theory based on conic sections. Our approach is algebraic: for a given real polynomial $P(u)$, the complete trigonometric system for the curve $x^2P(y/x) = 1$ is shown to be governed by the quotient ring $\mathbb{R}[u]/(P(u))$. The central result is a Factorization-Dynamics Correspondence, established via the Chinese Remainder Theorem. We prove that the factorization of $P(u)$ into linear, irreducible quadratic, and repeated factors directly determines the analytic form of the angle function and the qualitative dynamics of the parametrization. Specifically, linear factors yield logarithmic terms corresponding to dilational motion, irreducible quadratics yield arctangent terms for rotational motion, and repeated factors introduce nilpotent algebraic structures that manifest as rational, non-transcendental corrections to the angle. The framework is developed in detail for the hexagonal case ($x^2 - xy + y^2 = 1$), for which we derive the angle function, trigonometric identities, and governing ODE as consequences of the ring structure. The predictive power of the correspondence is further illustrated with three case studies beyond conics: a purely rotational system on a lemniscate-type quartic ($P(u) = 1 + u^4$), an algebraic spiral ($P(u) = 1 + u^3$), and a nilpotent-modified rotation on a circle ($P(u) = (1 + u^2)^2$). This work provides a systematic blueprint for generating new transcendental functions and their identities, positioning an algebra-first methodology in contrast to metric-based generalizations of complex numbers.

1 Introduction

The relationship between the complex numbers $\mathbb{C} = \mathbb{R}[i]/(i^2 + 1)$ and circular trigonometry is the archetype for a rich family of correspondences between algebra and geometry. This principle was extended to all conic sections by Harkin and Harkin [3], who demonstrated that their respective trigonometries are governed by two-dimensional real algebras with elliptic, parabolic, or hyperbolic signatures, corresponding to the complex, dual, and split-complex numbers. This work connects directly to the broader theory of hypercomplex numbers, where, for instance, the hyperbolic numbers ($u^2 = 1$) provide the algebraic foundation for the geometry of Minkowski spacetime [4]. This paper poses a natural next question: can this

paradigm be lifted from a *metric-quadratic* setting to a general *polynomial-algebraic* one? This inquiry resolves into two guiding questions:

(a) What constitutes a consistent and complete trigonometry when the 'unit locus' is defined by the curve $x^2P(y/x) = 1$ for an arbitrary real polynomial P?

(b) Which algebra replaces the complex numbers to govern the angle measure, exponential law, and identities for this generalized system?

Our answer is that the quotient ring $A = \mathbb{R}[u]/(P(u))$ provides the complete algebraic foundation. The structure of this ring, as revealed by the Chinese Remainder Theorem, directly controls the analytic form of the angle parameter $t(u) = \int 1/P$ and, through it, the dynamics of the parametrization and the entire system of identities flowing from an Euler-type element E_P. This algebra-first approach differs fundamentally from metric generalizations, such as the l_p-geometries of Richter [6] or the elliptical variables of Özen [5]. In our framework, the method of constructing a trigonometry remains classical and uniform; it is the underlying algebra that we generalize, allowing the choice of any polynomial P to define a new, consistent trigonometric system. The system's identities are derived via two complementary engines: algebraically, from the component expansion of the exponential element $E_P(t) = e^{tu}$, and geometrically, via a duplication construction that yields a powerful identity for the generalized tangent and a universal sector area law.

As a primary archetype, we develop the trigonometry for the hexagonal locus $x^2-xy+y^2 = 1$. We summarize the hexagonal case outcomes compactly: we derive an explicit angle function

$$t(u) = \frac{2}{\sqrt{3}}\left[\arctan\frac{2u-1}{\sqrt{3}} + \frac{\pi}{6}\right], \tag{1}$$

its inversion $u(t)$, and closed forms for the trigonometric pair $(\cos_P, \sin_P)$. From the Euler element

$$E_P(t) = \cos_P(t) + j\sin_P(t), \tag{2}$$

we generate a complete identity sheet (Pythagorean, sum, double-angle) and the governing linear ODE: $(\cos_P', \sin_P') = (\frac{1}{2}\cos_P - \sin_P,\ \cos_P - \frac{1}{2}\sin_P)$. We note that this algebra is isomorphic to the complex numbers, $\mathbb{H} \cong \mathbb{C}$ (as real algebras), and its norm is precisely the defining quadratic form, $N(x + jy) = x^2 - xy + y^2$.

The hexagonal case illustrates a general principle, which we formalize as the Factorization-Dynamics Correspondence. This correspondence, stated precisely in Theorem 3.8, shows how the partial fraction decomposition of $1/P$ mirrors the CRT factors of the ring $\mathbb{R}[u]/(P(u))$. This provides a structural dictionary connecting algebraic factors to geometric dynamics. We then view three case studies that test this blueprint beyond the familiar conic sections: a periodic system from $P(u) = 1 + u^4$, a spiral system from $P(u) = 1 + u^3$, and a nilpotent-modified rotational system from $P(u) = (1 + u^2)^2$.

The roadmap for this paper is as follows: §2 provides a complete, step-by-step derivation of the hexagonal archetype; §3 introduces the general blueprint and the main correspondence theorem; §4 develops the three case studies in full detail; §5 provides a synthesis and places the work within the existing literature.

2 The Hexagonal Case as Archetype

2.1 A Motivating Identity

Classical trigonometry is founded on the Pythagorean identity $\cos^2 t + \sin^2 t = 1$, which defines the unit circle as the fundamental locus. The natural evolution of this idea is to consider other quadratic forms. Consider, for instance, the equation

$$x^2 - xy + y^2 = 1.$$

This equation describes a tilted ellipse, centered at the origin. Its tilted nature can be seen by applying a coordinate rotation. If we let $x = X\cos\phi - Y\sin\phi$ and $y = X\sin\phi + Y\cos\phi$, the cross-term xy vanishes when $\phi = \pi/4$. In this rotated frame, the equation becomes $\frac{1}{2}X^2 + \frac{3}{2}Y^2 = 1$, the standard form for an ellipse. This simple observation confirms that non-Pythagorean quadratic forms can define familiar geometric shapes.

2.2 The Algebraic Frame: Hexagonal Numbers

The appropriate algebraic setting for our tilted ellipse is the ring of *hexagonal numbers*, a name derived from the fact that its integer-coefficient sublattice forms the Eisenstein integers, $\mathbb{Z}[e^{2\pi i/3}]$, which tile the plane in a hexagonal grid. This two-dimensional real algebra is defined by the quotient ring:

$$\mathbb{H} := \mathbb{R}[j]/(j^2 + j + 1).$$

This is the set of numbers of the form $x + jy$, where $x, y \in \mathbb{R}$ and the imaginary unit j satisfies the rule $j^2 = -1 - j$. This algebra possesses a multiplicative norm N. To derive it, we find the matrix representation for an element $z = x + jy$ by considering its multiplicative action on the basis $\{1, j\}$:

$$\begin{aligned} z \cdot 1 &= x + jy \\ z \cdot j &= xj + yj^2 = xj + y(-1 - j) = -y + (x - y)j. \end{aligned}$$

This linear transformation corresponds to the matrix $M_z = \begin{pmatrix} x & -y \\ y & x - y \end{pmatrix}$. The norm is the determinant of this matrix:

$$N(x + jy) = \det(M_z) = x(x - y) - (-y)(y) = x^2 - xy + y^2. \tag{3}$$

Thus, the unit locus of the algebra $\mathbb{H}$, defined by $N(z) = 1$, is precisely our target ellipse. The algebra $\mathbb{H}$ is isomorphic to $\mathbb{C}$ (as real algebras) via the map $\phi : \mathbb{H} \to \mathbb{C}$ defined by $\phi(x + jy) = x + ye^{2\pi i/3}$. To confirm this is an isomorphism, we check that ϕ preserves the algebraic structure. Specifically, $\phi(j) = e^{2\pi i/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ must satisfy the defining relation for j:

$$(\phi(j))^2 + \phi(j) + 1 = \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^2 + \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) + 1 = \left(\frac{1}{4} - i\frac{\sqrt{3}}{2} - \frac{3}{4}\right) - \frac{1}{2} + i\frac{\sqrt{3}}{2} + 1 = 0.$$

This places $\mathbb{H}$ in the "elliptic" class of generalized complex numbers, alongside $\mathbb{C}$.

2.3 Constructing the Trigonometric Functions

To define the trigonometric functions, we first require a notion of angle. Using slope coordinates $u = y/x$, we can write the equation of the unit ellipse as $x^2(1 - u + u^2) = 1$. Let $P(u) = u^2 - u + 1$. The angle parameter, t, is defined via a rational integral that effectively measures the "area" swept out in a particular sense:

$$t(u) = \int_0^u \frac{\mathrm{d}w}{P(w)} = \int_0^u \frac{\mathrm{d}w}{w^2 - w + 1}. \tag{4}$$

To evaluate this integral, we complete the square in the denominator:

$$t(u) = \int_0^u \frac{\mathrm{d}w}{(w - \frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} = \left[\frac{2}{\sqrt{3}} \arctan \frac{2w - 1}{\sqrt{3}}\right]_0^u = \frac{2}{\sqrt{3}} \left[\arctan \frac{2u - 1}{\sqrt{3}} + \frac{\pi}{6}\right].$$

This angle function is monotonically increasing and thus invertible; we write $u(t) = t^{-1}(u)$ for the slope as a function of angle parameter t. Solving for u:

$$\frac{\sqrt{3}}{2}t - \frac{\pi}{6} = \arctan\left(\frac{2u - 1}{\sqrt{3}}\right) \implies u(t) = \frac{1}{2} + \frac{\sqrt{3}}{2} \tan\left(\frac{\sqrt{3}}{2}t - \frac{\pi}{6}\right).$$

Our parametrization is given by $x = 1/\sqrt{P(u)}$ and $y = u/\sqrt{P(u)}$. The denominator is:

$$P(u(t)) = \left(u(t) - \frac{1}{2}\right)^2 + \frac{3}{4} = \frac{3}{4} \tan^2\left(\frac{\sqrt{3}}{2}t - \frac{\pi}{6}\right) + \frac{3}{4} = \frac{3}{4} \sec^2\left(\frac{\sqrt{3}}{2}t - \frac{\pi}{6}\right).$$

Substituting this into the parametrization and simplifying using trigonometric identities defines the hexagonal cosine and sine functions.

Definition 2.1 (Hexagonal Trigonometric Functions)**.** The hexagonal trigonometric functions, $\cos_P(t)$ and $\sin_P(t)$, are defined as:

$$\begin{aligned}
\cos_P(t) &:= \cos\left(\frac{\sqrt{3}}{2}t\right) + \frac{1}{\sqrt{3}} \sin\left(\frac{\sqrt{3}}{2}t\right) \\
\sin_P(t) &:= \frac{2}{\sqrt{3}} \sin\left(\frac{\sqrt{3}}{2}t\right)
\end{aligned}$$

These explicit forms are derived by substituting $u(t)$ into the parametrization formulas and applying angle subtraction identities. For instance, the derivation for $\cos_P(t)$ proceeds as:

$$\cos_P(t) = \frac{1}{\sqrt{P(u(t))}} = \frac{1}{\sqrt{\frac{3}{4} \sec^2\left(\frac{\sqrt{3}}{2}t - \frac{\pi}{6}\right)}} = \frac{2}{\sqrt{3}} \cos\left(\frac{\sqrt{3}}{2}t - \frac{\pi}{6}\right).$$

which simplifies to the expression in the definition.

2.4 The Hexagonal Identity Calculus

The true power of this construction is revealed through the exponential element:

$$E_P(t) = \cos_P(t) + j \sin_P(t) \in \mathbb{H}. \tag{5}$$

This element is the hexagonal analogue of Euler's formula $e^{it} = \cos t + i \sin t$. We can prove that it behaves like a classical exponential by applying the isomorphism ϕ:

$$\begin{aligned}\phi(E_P(t)) &= \phi\left(\left(\cos(\tfrac{\sqrt{3}}{2}t) + \frac{1}{\sqrt{3}}\sin(\tfrac{\sqrt{3}}{2}t)\right) + j\left(\frac{2}{\sqrt{3}}\sin(\tfrac{\sqrt{3}}{2}t)\right)\right)\\ &= \cos(\tfrac{\sqrt{3}}{2}t) + i\sin(\tfrac{\sqrt{3}}{2}t) = e^{i(\sqrt{3}/2)t}.\end{aligned}$$

Since ϕ is a ring isomorphism, it preserves multiplication. The exponential law $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$ in $\mathbb{C}$ implies an analogous law in $\mathbb{H}$, $E_P(t+s) = E_P(t)E_P(s)$. By expanding this product in $\mathbb{H}$ and equating components, we can derive a set of trigonometric identities.

$$\begin{aligned}E_P(t)E_P(s) &= (\cos_P(t) + j\sin_P(t))(\cos_P(s) + j\sin_P(s))\\ &= \cos_P(t)\cos_P(s) + j(\cos_P(t)\sin_P(s) + \sin_P(t)\cos_P(s)) + j^2\sin_P(t)\sin_P(s)\\ &= \cos_P(t)\cos_P(s) - \sin_P(t)\sin_P(s) + j(\cos_P(t)\sin_P(s) + \sin_P(t)\cos_P(s) - \sin_P(t)\sin_P(s)).\end{aligned}$$

Equating this with $E_P(t+s) = \cos_P(t+s) + j\sin_P(t+s)$ yields the sum formulas.

Hexagonal Identity Sheet

- **Pythagorean Identity:** $\cos_P^2(t) - \cos_P(t)\sin_P(t) + \sin_P^2(t) = 1$ (from the norm, Eq. 3)
- **Sum Formulas:**

$$\begin{aligned}\cos_P(t+s) &= \cos_P(t)\cos_P(s) - \sin_P(t)\sin_P(s)\\ \sin_P(t+s) &= \cos_P(t)\sin_P(s) + \sin_P(t)\cos_P(s) - \sin_P(t)\sin_P(s)\end{aligned}$$

- **Double Angle Formulas ($s = t$):**

$$\begin{aligned}\cos_P(2t) &= \cos_P^2(t) - \sin_P^2(t)\\ \sin_P(2t) &= 2\cos_P(t)\sin_P(t) - \sin_P^2(t)\end{aligned}$$

We find the derivative of $E_P(t)$ using its image under ϕ.

$$\phi(E_P'(t)) = \frac{d}{dt}\phi(E_P(t)) = \frac{d}{dt}e^{i(\sqrt{3}/2)t} = i\frac{\sqrt{3}}{2}e^{i(\sqrt{3}/2)t} = i\frac{\sqrt{3}}{2}\phi(E_P(t)).$$

The element in $\mathbb{H}$ that maps to $i\frac{\sqrt{3}}{2}$ is $k = \frac{1}{2} + j$. Thus, $E'_P(t) = kE_P(t) = (\frac{1}{2} + j)(\cos_P(t) + j\sin_P(t))$. Expanding this gives:

$$\begin{aligned}\cos'_P(t) + j\sin'_P(t) &= \frac{1}{2}\cos_P(t) + j\cos_P(t) + \frac{1}{2}j\sin_P(t) + j^2\sin_P(t)\\ &= \left(\frac{1}{2}\cos_P(t) - \sin_P(t)\right) + j\left(\cos_P(t) - \frac{1}{2}\sin_P(t)\right).\end{aligned}$$

Equating components gives the derivative rules:

$$\begin{aligned}\frac{d}{dt}\cos_P(t) &= \frac{1}{2}\cos_P(t) - \sin_P(t)\\ \frac{d}{dt}\sin_P(t) &= \cos_P(t) - \frac{1}{2}\sin_P(t)\end{aligned} \tag{6}$$

3 The General Blueprint and Its Structure

3.1 From Hexagons to Arbitrary Polynomials

The detailed procedure for the hexagonal case suggests a general method for constructing a trigonometry from any real polynomial.

Definition 3.1 (Polynomial Trigonometry)**.** Let $P(u)$ be a real polynomial such that the integral of its reciprocal is well-defined.

(a) The **unit locus** is the algebraic curve defined by the equation $x^2P(y/x) = 1$.

(b) The **angle parameter** t is defined via the slope u as $t(u) = \int_0^u \frac{\mathrm{d}w}{P(w)}$.

(c) Where $t(u)$ is invertible, let the inverse function be $u(t)$. The **trigonometric pair** $(\cos_P(t), \sin_P(t))$ is defined as:

$$\cos_P(t) = \frac{1}{\sqrt{P(u(t))}}, \qquad \sin_P(t) = \frac{u(t)}{\sqrt{P(u(t))}}.$$

By construction, these functions satisfy $(\cos_P(t))^2P(\sin_P(t)/\cos_P(t)) = 1$.

3.2 Exponential Element and Algebraic Identities

The identities of a P-trigonometry are generated by the exponential element in the associated quotient algebra.

Definition 3.2 (Exponential Element)**.** Let $A = \mathbb{R}[u]/(P(u))$ be the quotient algebra for a polynomial $P(u)$ of degree n. Let $\{1, u, \ldots, u^{n-1}\}$ be a basis for A. The **exponential element** is

$$E_P(t) = e^{tu} = \sum_{k=0}^{\infty}\frac{(tu)^k}{k!} = \sum_{k=0}^{n-1} s_k(t)u^k.$$

The functions $s_k(t)$ are the **generalized trigonometric components**.

Proposition 3.3 (Identity Generation)**.** The fundamental identities of the P-trigonometry are consequences of the ring multiplication in A:

(a) **Addition Formulas:** The components of $E_P(t+s) = E_P(t)E_P(s)$ yield the addition laws for $s_k(t+s)$.

(b) **Double-Angle Formulas:** The components of $E_P(2t) = (E_P(t))^2$ yield the double-angle laws for $s_k(2t)$.

(c) **Norm/Constraint Identity:** The defining relation $x^2P(y/x) = 1$ is the geometric expression of the norm induced by the algebra A.

Remark 3.4 (Nilpotent Truncation)**.** If $P(u)$ has repeated roots, the algebra A contains nilpotent elements. In such cases, the power series for $E_P(t)$ often truncates in certain algebraic directions, and identities acquire rational or polynomial-in-t corrections rather than being purely transcendental.

3.3 Geometric Duplication and Sector Area

A complementary geometric construction provides a double-angle law valid for any P-trigonometry and a universal formula for the sector area, calibrating the geometric meaning of the parameter t.

Theorem 3.5 (Geometric Duplication)**.** Let $\tan_P(t) = u(t)$. For any interval where the parametrization is well-defined,

$$\tan_P(2t) = \frac{\tan_P(t) + y(t)}{1 - y(t)\tan_P(t)},$$

where the correction term $y(t)$ is given by

$$y(t) = \tan\left(\int_t^{2t} \frac{P(u(s))}{1+u(s)^2}\mathrm{d}s\right).$$

Proof. This identity is derived from a geometric construction analogous to the classical proof for $\tan(\alpha+\beta)$. We consider the Euclidean angle ϕ that the position vector $(x(t), y(t))$ makes with the positive x-axis. By definition, $\tan\phi(t) = y(t)/x(t) = u(t) = \tan_P(t)$. The duplication law for $\tan_P(2t)$ can be seen as the tangent addition formula for the angles $\phi(t)$ and the difference angle $\Delta\phi = \phi(2t) - \phi(t)$. The term $y(t)$ in the theorem is precisely $\tan(\Delta\phi)$.

To find an expression for $\Delta\phi$, we compute the rate of change of the Euclidean angle ϕ with respect to the parameter t.

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\arctan\left(\frac{y(t)}{x(t)}\right) = \frac{1}{1+(y/x)^2}\frac{xy' - yx'}{x^2} = \frac{xy' - yx'}{x^2+y^2}.$$

From the definitions $x(t) = P(u(t))^{-1/2}$ and $y(t) = u(t)P(u(t))^{-1/2}$, and the fundamental ODE $u'(t) = P(u(t))$, we compute the derivatives using the chain rule:

$$x'(t) = -\frac{1}{2}P(u)^{-3/2}P'(u)u'(t) = -\frac{1}{2}x^3P'(u)P(u)$$
$$y'(t) = u'(t)x(t) + u(t)x'(t) = P(u)x(t) - \frac{1}{2}u(t)x^3P'(u)P(u).$$

Substituting these into the numerator of $\mathrm{d}\phi/\mathrm{d}t$:

$$\begin{aligned} xy' - yx' &= x(P(u)x - \frac{1}{2}ux^3P'(u)P(u)) - (ux)(-\frac{1}{2}x^3P'(u)P(u)) \\ &= x^2P(u) - \frac{1}{2}ux^4P'(u)P(u) + \frac{1}{2}ux^4P'(u)P(u) = x^2P(u). \end{aligned}$$

The denominator is $x^2 + y^2 = x^2(1 + u^2)$. Therefore,

$$\frac{\mathrm{d}\phi}{\mathrm{d}t} = \frac{x^2P(u)}{x^2(1+u^2)} = \frac{P(u(t))}{1+u(t)^2}.$$

The angle difference $\Delta\phi = \phi(2t) - \phi(t)$ is the integral of this expression from t to $2t$. Thus, $y(t) = \tan(\int_t^{2t} \frac{P(u(s))}{1+u(s)^2}\mathrm{d}s)$, and the theorem follows from the tangent addition formula. □

Theorem 3.6 (Sector Area)**.** For $t_0 \leq t_1$ in an interval where the parametrization is well-defined, the Euclidean area of the sector bounded by the unit curve $x^2P(y/x) = 1$ and the rays from the origin at parameters t_0 and t_1 is given by

$$A = \frac{t_1 - t_0}{2}.$$

Proof. The area of a sector in polar coordinates is given by the integral $A = \frac{1}{2}\int r^2\mathrm{d}\phi$, where ϕ is the polar angle. From the equation of the curve, $x^2P(y/x) = 1$, we substitute $x = r\cos\phi$ and $y = r\sin\phi$. This gives $r^2\cos^2\phi P(\tan\phi) = 1$, which implies $r^2 = \sec^2\phi/P(\tan\phi)$. The differential of the angle is $\mathrm{d}\phi$. We can relate this to the parameter t by changing variables. Let $u = \tan\phi$, so $\mathrm{d}u = \sec^2\phi\mathrm{d}\phi$, which means $\mathrm{d}\phi = \frac{\mathrm{d}u}{\sec^2\phi}$. Substituting these into the area integral:

$$A = \frac{1}{2}\int_{\phi(t_0)}^{\phi(t_1)} \frac{\sec^2\phi}{P(\tan\phi)}\mathrm{d}\phi = \frac{1}{2}\int_{u(t_0)}^{u(t_1)} \frac{\sec^2\phi}{P(u)}\frac{\mathrm{d}u}{\sec^2\phi} = \frac{1}{2}\int_{u(t_0)}^{u(t_1)} \frac{\mathrm{d}u}{P(u)}.$$

By the definition of our parameter, this final integral is simply $\frac{1}{2}(t(u(t_1)) - t(u(t_0))) = \frac{1}{2}(t_1 - t_0)$. This remarkable result shows that the parameter t is universally proportional to the Euclidean area swept out by the parametrization, regardless of the choice of polynomial P. □

3.4 The Algebraic Structure via the Chinese Remainder Theorem

The qualitative nature of the resulting trigonometry is determined entirely by the algebraic structure of the quotient ring $\mathcal{A} = \mathbb{R}[u]/(P(u))$. This structure is revealed by the Chinese Remainder Theorem (CRT). Any real polynomial $P(u)$ can be factored into powers of distinct irreducible real polynomials:

$$P(u) = c \cdot P_1(u)^{m_1} P_2(u)^{m_2} \cdots P_k(u)^{m_k}.$$

Since the ideals $\langle P_i(u)^{m_i} \rangle$ are pairwise coprime, the CRT provides a fundamental ring isomorphism:

$$\mathcal{A} \cong \mathbb{R}[u]/\langle P_1(u)^{m_1} \rangle \times \cdots \times \mathbb{R}[u]/\langle P_k(u)^{m_k} \rangle.$$

This decomposition is the structural blueprint for the trigonometry.

Proposition 3.7 (Structure of Component Algebras)**.** The component rings $\mathcal{A}_i = \mathbb{R}[u]/\langle P_i(u)^{m_i} \rangle$ have the following structure:

(a) If $P_i(u) = u - a$ is a linear factor, $\mathcal{A}_i \cong \mathbb{R}[\varepsilon]/\langle \varepsilon^{m_i} \rangle$.

(b) If $P_i(u) = u^2 + Au + B$ is an irreducible quadratic, $\mathcal{A}_i \cong \mathbb{C}[\varepsilon]/\langle \varepsilon^{m_i} \rangle$.

Theorem 3.8 (The Factorization-Dynamics Correspondence)**.** The analytic form of the angle parameter $t(u) = \int 1/P$ and the dynamics of $(\cos_P(t), \sin_P(t))$ are determined by the component algebras of $\mathbb{R}[u]/(P(u))$ as follows:

(a) Semisimple Components ($m_i = 1$):

  (a) A factor $\mathbb{R}[u]/(u - a) \cong \mathbb{R}$ corresponds to a term $\dfrac{A}{u - a}$ in the partial fraction decomposition of $1/P$. Its integral is $A \ln |u - a|$, generating dilational (hyperbolic) dynamics, as seen in the trigonometry of Minkowski spacetime.

  (b) A factor $\mathbb{R}[u]/(u^2 + Au + B) \cong \mathbb{C}$ corresponds to a term $\dfrac{Bu + C}{u^2 + Au + B}$. Its integral is a sum of a logarithmic and an arctangent term, generating rotational (periodic) dynamics.

(b) Nilpotent Components ($m_i > 1$):

  (a) A factor $\mathbb{R}[\varepsilon]/\langle \varepsilon^{m_i} \rangle$ or $\mathbb{C}[\varepsilon]/\langle \varepsilon^{m_i} \rangle$ corresponds to terms like $\dfrac{A}{(u - a)^k}$ or $\dfrac{Bu + C}{(u^2 + Au + B)^k}$ with $k > 1$. Their integrals are rational algebraic functions that modify the base dynamics.

This theorem establishes a precise structural dictionary: $\mathbb{R}$ factors mean logarithms and scaling; $\mathbb{C}$ factors mean arctangents and rotation; and repeated factors (nilpotents) mean additional algebraic terms.

4 Case Studies Beyond Conic Sections

For each case, we identify the algebra and its CRT decomposition, state the angle function, and present key identities derived from the algebraic component method and the geometric duplication law.

4.1 Case 1: Lemniscate-Type Quartic ($P(u) = 1 + u^4$)

The algebra is $\mathcal{A} = \mathbb{R}[u]/(u^4 + 1) \cong \mathbb{C} \times \mathbb{C}$, predicting purely periodic, rotational dynamics. The unit locus is $x^4 + y^4 = x^2$. The angle parameter, from the partial fraction decomposition of $1/(1 + w^4)$, is

$$t(u) = \frac{1}{4\sqrt{2}} \ln \frac{u^2 + \sqrt{2}u + 1}{u^2 - \sqrt{2}u + 1} + \frac{\sqrt{2}}{4} \left(\arctan(\sqrt{2}u + 1) + \arctan(\sqrt{2}u - 1) \right).$$

The geometric consequence for the generalized tangent is the rational duplication law:

$$\tan_P(2t) = \frac{2 \tan_P(t)(1 - \tan_P(t)^2)}{1 + 2 \tan_P(t)^2 - \tan_P(t)^4}.$$

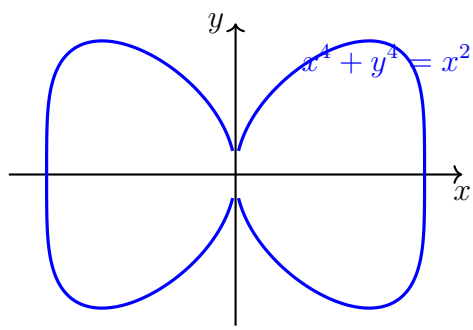

Figure 1: The lemniscate-type curve generated by $P(u) = 1 + u^4$.

4.2 Case 2: An Algebraic Spiral ($P(u) = 1 + u^3$)

The algebra is $\mathcal{A} = \mathbb{R}[u]/(u^3 + 1) \cong \mathbb{R} \times \mathbb{C}$, predicting mixed dilational and rotational dynamics, i.e., a spiral. The unit locus is the cubic curve $x^3 + y^3 = x$. The angle function reflects the mixed algebra:

$$t(u) = \frac{1}{6} \ln \frac{(u + 1)^2}{u^2 - u + 1} + \frac{1}{\sqrt{3}} \arctan \left(\frac{2u - 1}{\sqrt{3}} \right) + \frac{\pi}{6\sqrt{3}}.$$

The identities are governed by the algebra with reduction rule $u^3 = -1$. The exponential element is $E_P(t) = C_0(t) + C_1(t)u + C_2(t)u^2$. Squaring this element and reducing with $u^3 = -1$ and $u^4 = -u$:

$$\begin{aligned} E_P(t)^2 &= (C_0 + C_1 u + C_2 u^2)^2 = C_0^2 + 2C_0C_1 u + (C_1^2 + 2C_0C_2)u^2 + 2C_1C_2u^3 + C_2^2u^4 \\ &= (C_0^2 - 2C_1C_2) + (2C_0C_1 - C_2^2)u + (C_1^2 + 2C_0C_2)u^2. \end{aligned}$$

This gives the component identities, e.g., $C_0(2t) = C_0(t)^2 - 2C_1(t)C_2(t)$.

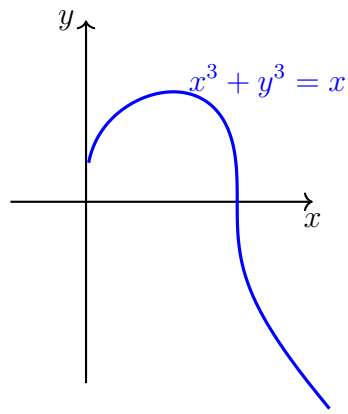

Figure 2: The algebraic spiral from $P(u) = 1 + u^3$, covering the primary branch for $u > -1$.

4.3 Case 3: Nilpotent-Modified Rotation ($P(u) = (1 + u^2)^2$)

The algebra is $\mathcal{A} = \mathbb{R}[u]/\langle(u^2 + 1)^2\rangle \cong \mathbb{C}[\varepsilon]/\langle\varepsilon^2\rangle$, where $\varepsilon = u^2 + 1$ is a nilpotent element. This structure predicts a base rotational dynamic modified by a non-transcendental, algebraic term. The unit locus is the circle $(x - 1/2)^2 + y^2 = 1/4$. The angle function confirms this structure:

$$t(u) = \frac{1}{2}\arctan(u) + \frac{u}{2(1 + u^2)}.$$

The first term is purely rotational, while the second is a rational correction from the nilpotent component. The identities are governed by the algebra with reduction rule $u^4 = -2u^2 - 1$. The presence of the nilpotent element fundamentally alters the nature of these identities. The structure of the exponential is best understood in the $\mathbb{C}[\varepsilon]$ basis as being proportional to $e^{i\omega t}(1 + kt\varepsilon)$ for some complex constants ω and k. This form, a rotation modulated by a linear growth term in the nilpotent direction, does not translate to a simple rational addition law for the slope $\tan_P(t)$.

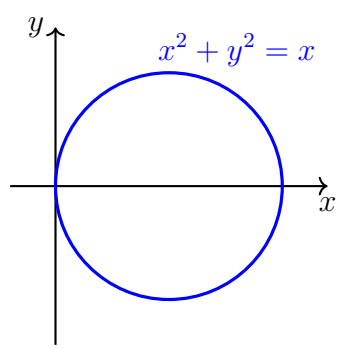

Figure 3: The circular locus generated by the repeated-root polynomial $P(u) = (1 + u^2)^2$.

5 Conclusion

The central principle of this paper is that the algebraic structure of the quotient ring $\mathbb{R}[u]/(P(u))$ provides a complete blueprint for the trigonometry of the planar curve defined by $x^2P(y/x) = 1$. By moving from a metric-based view to an algebra-first perspective, we have generalized the well-known correspondence between conic sections and two-dimensional hypercomplex numbers to a framework applicable to any real polynomial. The main theoretical result, the Factorization-Dynamics Correspondence, establishes a precise structural dictionary: the decomposition of the ring via the Chinese Remainder Theorem directly predicts the analytic form of the angle parameter $t(u) = \int 1/P$ and the qualitative dynamics of the resulting parametrization. Identity generation is not ad hoc: it is the shadow of the exponential in $A = \mathbb{R}[u]/(P(u))$ and its CRT decomposition, complemented by a geometric duplication law and a universal sector-area law. Semisimple factors yield classical trigonometric channels; repeated factors yield nilpotent corrections.

This work builds upon the foundational classification of trigonometries for quadratic forms established by Harkin and Harkin [3]. Their results for the elliptic, parabolic, and hyperbolic cases are subsumed in our framework as the semisimple slice where $\deg P = 2$. Our approach is also compatible with the broader hypercomplex frameworks developed by Catoni et al. [1, 4], but it is distinguished by its direct computational pipeline, which proceeds from the choice of a single polynomial P to the automatic generation of an angle function, an Euler-type exponential element, and a complete set of identities derived from the ring's multiplication law. Furthermore, our analysis of nilpotent components, which arise from repeated factors of P, extends the specific study of dual-number dynamics, such as that explored by Dündar [2], to the full class of non-semisimple algebras $\mathbb{R}[\varepsilon]/(\varepsilon^m)$ and $\mathbb{C}[\varepsilon]/(\varepsilon^m)$. This algebraic generalization stands in contrast to metric-focused approaches, such as the l_p-complex numbers of Richter [6], by generalizing the algebra itself while preserving the classical trigonometric construction. This ensures that an Euler law, $E_P(t+s) = E_P(t)E_P(s)$, provides all identities "for free."

The case studies presented herein confirmed the predictive power of this structural dictionary. The purely rotational dynamics of the lemniscate-type quartic emerged directly from the $\mathbb{C} \times \mathbb{C}$ algebra of $P(u) = 1 + u^4$. The spiral trajectory of the cubic curve was a direct consequence of the mixed $\mathbb{R} \times \mathbb{C}$ algebra of $P(u) = 1 + u^3$. Finally, the nilpotent-modified rotation on a circle arose from the $\mathbb{C}[\varepsilon]/(\varepsilon^2)$ algebra of $P(u) = (1 + u^2)^2$, with the angle function exhibiting the predicted rational correction. These examples show that complex geometric behaviors can be understood and predicted through the elementary factorization of a polynomial. This framework suggests several avenues for future research. A natural extension is to higher-dimensional analogues, exploring parametrizations of polynomial surfaces governed by algebras such as $\mathbb{R}[u_1, \ldots, u_k]/(P_1, \ldots, P_r)$. Another promising direction is to investigate the deeper geometric meaning of the nilpotent corrections, possibly connecting the linear growth term manifest in the exponential element $e^{i\omega t}(1 + kt\varepsilon)$ to concepts like curvature or infinitesimal deformations of the base curve.

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